# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 10 — SCIENCE · HARD LEVEL UNIT TEST SET 3

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 15 × 1 = 15

*Choose the correct option and write it in your answer book.*

**1.** A single Amoeba undergoes binary fission every 20 minutes. Starting with one Amoeba, the number after 1 hour is 1

**(a)** 6 **(b)** 16 **(c)** 8 **(d)** 3

**2.** A tall plant is crossed with a dwarf plant; 52 of 100 offspring are tall. The tall parent was 1

**(a)** Tt **(b)** TT

**(c)** tt **(d)** cannot be determined

**3.** Two potted plants are kept in closed chambers: one in the dark, the other in light with a dish of KOH solution (which absorbs carbon dioxide). Starch will be absent in: 1

**(a)** both plants **(b)** only the plant with KOH

**(c)** only the plant in the dark **(d)** neither plant

**4.** Overproduction of growth hormone in childhood causes: 1

**(a)** gigantism **(b)** diabetes **(c)** goitre **(d)** dwarfism

**5.** A woman's oviducts are blocked. Which would be affected most? 1

**(a)** The egg cannot meet the sperm **(b)** production of eggs

**(c)** production of sperm **(d)** menstruation of the male

**6.** A farmer saves seeds from tall plants raised in good soil. The next generation is grown in poor soil and is short. This shows 1

**(a)** blending

**(b)** environment affects expression of the trait

**(c)** tall is recessive

**(d)** genes were lost

**7.** Assertion (A): Xylem transports water upward in tall trees. Reason (R): Transpiration creates a suction force that pulls water. 1

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is true but R is false.

**(d)** A is false but R is true.

**8.** Assertion (A): Iodised salt is recommended for good health. Reason (R): Iodine is required for the synthesis of thyroxine. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**9.** Assertion (A): Cross-pollination leads to greater variation. Reason (R): It combines genes from two different plants. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**10.** Assertion (A): In a monohybrid cross the F2 genotypic ratio is 1 : 2 : 1. Reason (R): F1 plants form two kinds of gametes in equal numbers. 1

**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**11.** A heart pumps 70 mL of blood per beat at 72 beats per minute. Volume pumped in a minute (in mL) is: 1

**(a)** 7200 **(b)** 720 **(c)** 5040 **(d)** 504

**12.** An impulse is blocked in the sensory neuron only. The patient will: 1

**(a)** not feel a stimulus **(b)** feel but not move
**(c)** move more **(d)** have no reflex only

**13.** Starting with one yeast cell, after 4 successive buddings of each cell that each give one new cell (doubling), the number is 1

**(a)** 8 **(b)** 16 **(c)** 4 **(d)** 32

**14.** A plant of genotype TtRr produces gametes TR, Tr, tR and tr in equal numbers. The probability of a gamete tr is 1

**(a)** 1/4 **(b)** 1/8 **(c)** 1 **(d)** 1/2

**15.** Lime water turns milky faster in a test tube with exhaled air than in inhaled air because exhaled air has: 1

**(a)** less water **(b)** more nitrogen
**(c)** more oxygen **(d)** more carbon dioxide

## SECTION — B (Very Short Answer Type) 5 × 2 = 10

*Answer in about 20–30 words each.*

**16.** Name the structure that develops into the male gamete carrier in flowering plants and state where it is formed. 2

**17.** A man with attached earlobes (ff) and a woman with free (Ff) have 4 children. Expected number with attached earlobes? 2

**18.** The glomerular filtrate is 180 L per day and urine is 1.5 L. Find the volume of water reabsorbed and the percentage. 2

**19.** Why does the roots' growth against light (negative phototropism) benefit the plant? 2

**20.** Starting with one Amoeba and dividing every 30 minutes, find the number after 3 hours. 2

## SECTION — C (Short Answer Type) — 5 × 3 = 15

*Answer in about 40–60 words each.*

**21.** Explain why a woman who has her ovaries removed will not have menstrual cycles. 3

**22.** A couple has three sons. What is the probability that the fourth child is a girl? Explain how the answer is obtained. 3

**23.** A sample of urine shows glucose. Which part of the nephron is faulty? Explain the normal process. 3

**24.** How do plants ensure that the pollen tube reaches the ovule? 3

OR

A flower has a long style. What is the effect on the pollen tube and how does it reach the ovule? 3

**25.** In a population of beetles, a green colour variant arises through a DNA copying error. Birds eat red beetles more. Explain what happens to the proportion of green beetles over generations. 3

OR

Explain the absorption of water by the roots and the role of osmosis. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**26.** Calculate: a person breathes 15 times per minute with 500 mL per breath. Find the volume of air in an hour and the oxygen inhaled if air has 21% oxygen. 5

OR

A person touches a hot plate, quickly pulls the hand and then screams. Explain the sequence in terms of reflex arc and brain involvement with a flow chart. 5

**27.** A student sets up three petri dishes of bread: one moist and warm, one dry, one moist and cold. Predict the results and explain with respect to spores. 5

OR

A tall plant with round seeds (TtRr) is crossed with a dwarf plant with round seeds (ttRr). Find the expected phenotypic ratio of the offspring. 5

## END OF QUESTION PAPER — Total: 50 Marks